R SHEAFHOM(K, R SHEAFHOM(L,M)) = R SHEAFHOM(K OTIMES ^L L, M)

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Hom-tensor compose identity. Cohomology of Sheaves, Lemma cohomology-lemma-compose. SheafHom^*(K, SheafHom^*(L,I)) = SheafHom^*(Tot(K otimes L), I).
- Hom complex into K-injectives is K-injective. Cohomology of Sheaves, Lemma cohomology-lemma-RHom-from-K-flat-into-K-injective. Representation of the left-hand side.
- Sections of R SheafHom. Cohomology of Sheaves, Lemma cohomology-lemma-section-RHom-over-U. Global sections compute Hom in the derived category.

Source reference. Original equation 08DI: sources/stacks-chow/tags.txt, source lines 10638–10638. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0A8M: sources/stacks-chow/tags.txt, source lines 12920–12920. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0A8T: sources/stacks-chow/tags.txt, source lines 12926–12926. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08DK: sources/stacks-chow/tags.txt, source lines 10640–10640. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let K, L, M be objects of $D(\mathcal{O}_X)$. With the construction as described above there is a canonical isomorphism*

$$R\mathcal{H}om(K, R\mathcal{H}om(L, M)) = R\mathcal{H}om(K \otimes^{\mathbf{L}}_{\mathcal{O}_X} L, M)$$

in $D(\mathcal{O}_X)$ functorial in K, L, M which recovers (08DI) by taking $H^0(X, -)$.

Proof. Choose a K-injective complex $\mathcal{I}^\bullet$ representing M and a K-flat complex of $\mathcal{O}_X$-modules $\mathcal{L}^\bullet$ representing L. Let $\mathcal{K}^\bullet$ be any complex of $\mathcal{O}_X$-modules representing K. Then we have

$$\mathcal{H}om^\bullet(\mathcal{K}^\bullet, \mathcal{H}om^\bullet(\mathcal{L}^\bullet, \mathcal{I}^\bullet)) = \mathcal{H}om^\bullet(\mathrm{Tot}(\mathcal{K}^\bullet \otimes_{\mathcal{O}_X} \mathcal{L}^\bullet), \mathcal{I}^\bullet)$$

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by Lemma 0A8M. Note that the left hand side represents $R\mathcal{H}om(K, R\mathcal{H}om(L, M))$ (use Lemma 0A8T) and that the right hand side represents $R\mathcal{H}om(K \otimes^{\mathbf{L}}_{\mathcal{O}_X} L, M)$. This proves the displayed formula of the lemma. Taking global sections and using Lemma 08DK we obtain (08DI). $\square$